Arista Access Node MG-ACC-38 QoS Congestion Event SOP

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8118

Description

The Access Node MG-ACC-38, located at the MG Road Core site, is experiencing a Quality of Service (QoS) congestion event, as indicated by the ALARM_8118 warning. This is attributed to the occurrence of microbursts and interface tail drops, ultimately resulting in QoS queue starvation. This phenomenon is a direct consequence of the Arista EOS's queuing mechanism, specifically the 'fair-queue' configuration, which allocates bandwidth and buffer resources to each queue based on the 'bandwidth-share' and 'buffer-size' settings. When the configured bandwidth and buffer resources are exceeded, causing excessive packet drops, the queue becomes starved, leading to congestion. The 'show queue' and 'show interface' commands can be used to analyze the queue and interface statistics, respectively, thereby aiding in the identification of the root cause of the congestion event.

Pre-requisite Checks

- Ensure remote hands are dispatched to the MG Road Core site for assistance.
- Verify the status of the redundant links to ensure that they are operational.
- Obtain the current 'show run' configuration of the Access Node MG-ACC-38.

Potential Root Causes

Cause	Probability
Bandwidth Share Exceeded	High
Buffer Size Insufficient	Medium
QoS Policy Misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: show queue, Expected Output: Queue statistics, including queue ID, packet drops, and bandwidth allocation., Description: This command displays the current queue statistics, allowing for the identification of queues experiencing packet drops. This information can be used to correlate with the congestion event and pinpoint the root cause.

Command: show interface, Expected Output: Interface statistics, including packet drops, errors, and bandwidth utilization., Description: This command displays the current interface statistics, providing vital information regarding packet drops, errors, and bandwidth utilization. This can aid in the identification of the interface experiencing excessive packet drops, thereby contributing to the QoS congestion event.
```

```
Command: show qos policy, Expected Output: QoS policy configuration, including bandwidth shares and buffer sizes., Description: This command displays the current QoS policy configuration, allowing for the verification of the bandwidth shares and buffer sizes allocated to each queue. This information can be used to identify potential misconfigurations contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Increase Bandwidth Share, Command: configure terminal copy run start bandwidth-share , Description: Increase the bandwidth share allocated to the affected queue to alleviate congestion. This step involves modifying the 'bandwidth-share' setting to allocate more bandwidth to the queue, thereby reducing packet drops and QoS queue starvation.

Step 2: Step: Increase Buffer Size, Command: configure terminal copy run start buffer-size , Description: Increase the buffer size allocated to the affected queue to alleviate congestion. This step involves modifying the 'buffer-size' setting to increase the buffer resources available to the queue, thereby reducing packet drops and QoS queue starvation.

Step 3: Step: Reconfigure QoS Policy, Command: configure terminal copy run start qos policy bandwidth-share , Description: Reconfigure the QoS policy to allocate sufficient bandwidth and buffer resources to each queue. This step involves modifying the QoS policy configuration to ensure that each queue is allocated sufficient bandwidth and buffer resources, thereby preventing QoS queue starvation and congestion events.

Post-Check Verification

```
Command: show queue, Expected Output: Queue statistics, including queue ID, packet drops, and bandwidth allocation., Description: Verify that the queue statistics reflect a reduction in packet drops and an increase in bandwidth allocation after implementing the resolution steps.

Command: show interface, Expected Output: Interface statistics, including packet drops, errors, and bandwidth utilization., Description: Verify that the interface statistics reflect a reduction in packet drops and an increase in bandwidth utilization after implementing the resolution steps.

Command: show qos policy, Expected Output: QoS policy configuration, including bandwidth shares and buffer sizes., Description: Verify that the QoS policy configuration reflects the changes made to allocate sufficient bandwidth and buffer resources to each queue.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: configure terminal copy start run reload, Description: Restore the original configuration to undo any changes made during the resolution steps. This step involves using the 'copy start run' command to restore the original configuration and then reloading the Access Node MG-ACC-38 to ensure that the changes take effect.

- Step: Reapply Resolution Steps, Command: configure terminal copy run start bandwidth-share , Description: Reapply the resolution steps to re-establish the changes made to alleviate the congestion event. This step involves reapplying the 'bandwidth-share' and 'buffer-size' settings to ensure that each queue is allocated sufficient bandwidth and buffer resources.